Skills Summary

Direction, Form, and Strength in Scatter Plots

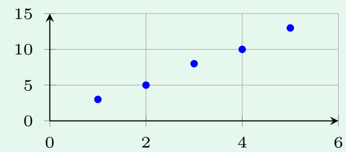
Use this reference while completing your worksheet or when studying.

Skill 1 — Direction: which way does the cloud go?

Ask: as you move right along the horizontal axis, do the points go *up* or *down*?

Up together → **positive** association. One up while the other goes down → **negative**. No tilt at all → **no association**. Always say it about the two real quantities, not just "up".

Example: Read the value at $x = 4$, then name the direction.



Step 1. Reading a value is a lookup; naming the direction means asking which way the points tilt.
Step 2. Above $x = 4$ the point sits level with 10, and the whole cloud rises left to right, so the direction is positive. ⇒ **10**, positive association

Try it: A plot has points $(1, 14)$, $(2, 11)$, $(3, 9)$, $(4, 6)$, $(5, 3)$. Read the value at $x = 3$ and name the direction.

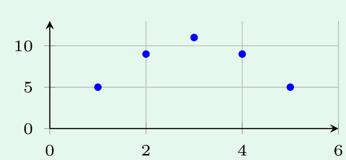
check: **9**, negative association

Skill 2 — Form: straight line or curve?

Ask: could one straight line follow *all* the points?

Yes → **linear**. No — the pattern rises then falls, or bends steeply → **non-linear**. A useful test: find the highest point. If it sits in the middle rather than at an end, the form cannot be linear.

Example: Find the input with the greatest output, then name the form.



Step 1. The form question is about shape, so locate the peak and see whether it sits at an end.
Step 2. The tallest point is above $x = 3$, in the middle, and the points fall away on both sides — an arch, not a line. ⇒ **3**, non-linear form

Try it: A plot has points $(1, 2)$, $(2, 5)$, $(3, 8)$, $(4, 11)$, $(5, 14)$. Find the input with the greatest output and name the form.

check: **5**, linear form

Skill 3 — Strength: how tightly do the points fit?

Ask: how far do the points stray from the pattern?

Hugging the line → **strong**. Scattered widely around it → **weak**. A quick numerical clue: for inputs close together, a small *range* of outputs means the points are packed tightly, a large range means they are spread out. A single far-away point is an **outlier** — describe it separately.

Example: Five points have outputs $10, 11, 12, 13, 14$. Find the range and say what it suggests about strength.
Step 1. Strength is about spread, so measure the spread of the outputs with the range.
Step 2. Largest 14, smallest 10, so the range is $14 - 10$ — a narrow band for five points.
⇒ **4**, a tightly packed, strong pattern

Try it: Five points have outputs $4, 15, 7, 13, 9$. Find the range and say what it suggests.

check: **11**, widely spread, so weak

(!) Watch out: direction, form and strength are three separate questions. A plot can be positive (direction) and still be non-linear (form) and weak (strength) — answer all three when you are asked to describe an association.